

112 Autumn

EEIE30076

系統晶片設計 SOC Design

Lab. #1

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系級：機械所碩一

一、 Brief introduction about the overall system

本實驗介紹 3 種軟體(Vitis HLS/Vivado/MobaXterm)，將 C++ 語言經過下列方法燒錄至 OnlineFPGA，並進行遠端驗證。

1. 透過 Vitis HLS 將 C++ 語言建置 IP
2. 透過 Vivado 將 IP 與 ZYNQ7 連結生成 .bit/.hwh 檔案
3. 透過 MobaXterm 登入 OnlineFPGA Server
4. Server 端開啟 Jupyter Notebook 連結
5. 將 .bit/.hwh/.ipynb 上傳至 Jupyter Notebook
6. 使用 Python 程式驗證 .bit/.hwh 是否正確

二、 What is observed & learned

1. Vitis

Vitis 中 Multiplication.cpp 僅有一個函式，其作用為將 n32In1*

n32In2 結果輸出至變數 pn32ResOut

```
void multip_2num(int32_t n32In1, int32_t n32In2, int32_t* pn32ResOut)
    *pn32ResOut = n32In1 * n32In2;
```

因此 Test Bench 之 MultipTester.cpp 調用函式 multip_2num 跑九九

乘法表，確認函式輸出結果是否與測試一致。

2. Multip2Num.ipynb

確認當前路徑

```
print(os.getcwd) # /home/xilinx/jupyter_notebooks
```

Python 程式會從當前路徑透過 ZYNQ Overlay 函式

開啟生成好的 Multip2Num.bit 。

```
ol = Overlay("/home/xilinx/jupyter_notebooks/Multip2Num.bit")
```

選用 Multip2Num.bit 當中”multip_2num_0”的 IP 。

```
regIP = ol.multip_2num_0
```

將 i, j 值寫入對應的 Control Address 裡面，並從對應的 Address

讀取數值回來。

```
regIP.write(0x10, i + 1)
```

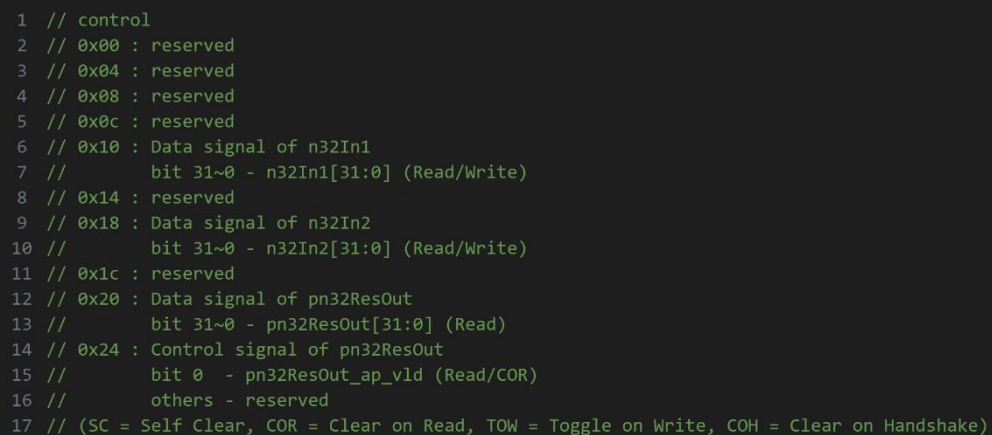
```
regIP.write(0x18, j + 1)
```

```
Res = regIP.read(0x20)
```

Address 之所以是 0x10, 0x18, 0x20，其原因

為”{ProjectDirectory}\{ProjectName}\solution1\impl\misc\drivers\mu

ltip_2num_v1_0\src\{Function}_hw.h”中有定義 Control Address 。



```
1 // control
2 // 0x00 : reserved
3 // 0x04 : reserved
4 // 0x08 : reserved
5 // 0x0c : reserved
6 // 0x10 : Data signal of n32In1
7 //      bit 31~0 - n32In1[31:0] (Read/Write)
8 // 0x14 : reserved
9 // 0x18 : Data signal of n32In2
10 //      bit 31~0 - n32In2[31:0] (Read/Write)
11 // 0x1c : reserved
12 // 0x20 : Data signal of pn32ResOut
13 //      bit 31~0 - pn32ResOut[31:0] (Read)
14 // 0x24 : Control signal of pn32ResOut
15 //      bit 0 - pn32ResOut_ap_vld (Read/COR)
16 //      others - reserved
17 // (SC = Self Clear, COR = Clear on Read, TOW = Toggle on Write, COH = Clear on Handshake)
```

三、 Screen dump

- Performance

```
1  =====
2  == Performance Estimates
3  =====
4  + Timing:
5      * Summary:
6      +-----+-----+-----+-----+
7      | Clock | Target | Estimated| Uncertainty|
8      +-----+-----+-----+-----+
9      | ap_clk | 10.00 ns| 6.912 ns| 2.70 ns|
10     +-----+-----+-----+-----+
11
12  + Latency:
13      * Summary:
14      +-----+-----+-----+-----+-----+-----+
15      | Latency (cycles) | Latency (absolute) | Interval | Pipeline|
16      | min | max | min | max | min | max | Type |
17      +-----+-----+-----+-----+-----+-----+
18      | 3 | 3 | 30.000 ns| 30.000 ns| 4 | 4 | no|
19      +-----+-----+-----+-----+-----+-----+
20
21  + Detail:
22      * Instance:
23      N/A
24
25      * Loop:
26      N/A
```

● Utilization

```
1 =====
2 -- Utilization Estimates
3 =====
4 * Summary:
5 +-----+-----+-----+-----+
6 |      Name      | BRAM_18K| DSP | FF | LUT | URAM|
7 +-----+-----+-----+-----+
8 | DSP            |         | -| -| -| -| -|
9 | Expression     |         | -| -| -| -| -|
10 | FIFO           |         | -| -| -| -| -|
11 | Instance       |         | 0| 3| 309| 282| -|
12 | Memory         |         | -| -| -| -| -|
13 | Multiplexer    |         | -| -| -| 25| -|
14 | Register       |         | -| -| 100| -| -|
15 +-----+-----+-----+-----+
16 | Total          |         | 0| 3| 409| 307| 0|
17 +-----+-----+-----+-----+
18 | Available      |         | 280| 220| 106400| 53200| 0|
19 +-----+-----+-----+-----+
20 | Utilization (%) |         | 0| 1| ~0| ~0| 0|
21 +-----+-----+-----+-----+
22
23 + Detail:
24 * Instance:
25 +-----+-----+-----+-----+
26 |      Instance      |      Module      | BRAM_18K| DSP| FF | LUT | URAM|
27 +-----+-----+-----+-----+
28 | control_s_axi_U    | control_s_axi    |         | 0| 0| 144| 232| 0|
29 | mul_32s_32s_32_2_1_U1 | mul_32s_32s_32_2_1 |         | 0| 3| 165| 50| 0|
30 +-----+-----+-----+-----+
31 | Total              |                  |         | 0| 3| 309| 282| 0|
32 +-----+-----+-----+-----+
33
34 * DSP:
35 N/A
36
37 * Memory:
38 N/A
39
40 * FIFO:
41 N/A
42
43 * Expression:
44 N/A
45
46 * Multiplexer:
47 +-----+-----+-----+-----+
48 |      Name      | LUT| Input Size| Bits| Total Bits|
49 +-----+-----+-----+-----+
50 | ap_NS_fsm      | 25|          5| 1| 5|
51 +-----+-----+-----+-----+
52 | Total          | 25|          5| 1| 5|
53 +-----+-----+-----+-----+
54
55 * Register:
56 +-----+-----+-----+-----+
57 |      Name      | FF | LUT| Bits| Const Bits|
58 +-----+-----+-----+-----+
59 | ap_CS_fsm      | 4| 0| 4| 0|
60 | mul_in1_reg_71 | 32| 0| 32| 0|
61 | n32In1_read_reg_66 | 32| 0| 32| 0|
62 | n32In2_read_reg_61 | 32| 0| 32| 0|
63 +-----+-----+-----+-----+
64 | Total          | 100| 0| 100| 0|
65 +-----+-----+-----+-----+
```

● Interface

```
1  =====
2  == Interface
3  =====
4  * Summary:
5  +-----+-----+-----+-----+-----+
6  |      RTL Ports      | Dir | Bits | Protocol | Source Object | C Type |
7  +-----+-----+-----+-----+-----+
8  | s_axi_control_AWVALID | in | 1 | s_axi | control | pointer |
9  | s_axi_control_AWREADY | out | 1 | s_axi | control | pointer |
10 | s_axi_control_AWADDR | in | 6 | s_axi | control | pointer |
11 | s_axi_control_WVALID | in | 1 | s_axi | control | pointer |
12 | s_axi_control_WREADY | out | 1 | s_axi | control | pointer |
13 | s_axi_control_WDATA | in | 32 | s_axi | control | pointer |
14 | s_axi_control_WSTRB | in | 4 | s_axi | control | pointer |
15 | s_axi_control_ARVALID | in | 1 | s_axi | control | pointer |
16 | s_axi_control_ARREADY | out | 1 | s_axi | control | pointer |
17 | s_axi_control_ARADDR | in | 6 | s_axi | control | pointer |
18 | s_axi_control_RVALID | out | 1 | s_axi | control | pointer |
19 | s_axi_control_RREADY | in | 1 | s_axi | control | pointer |
20 | s_axi_control_RDATA | out | 32 | s_axi | control | pointer |
21 | s_axi_control_RRESP | out | 2 | s_axi | control | pointer |
22 | s_axi_control_BVALID | out | 1 | s_axi | control | pointer |
23 | s_axi_control_BREADY | in | 1 | s_axi | control | pointer |
24 | s_axi_control_BRESP | out | 2 | s_axi | control | pointer |
25 | ap_clk | in | 1 | ap_ctrl_none | multip_2num | return value |
26 | ap_rst_n | in | 1 | ap_ctrl_none | multip_2num | return value |
27 +-----+-----+-----+-----+-----+
```

● Co-simulation transcript/waveform

Cosimulation Report for 'multip_2num'

General Information

Date: Tue Sep 19 14:44:21 2023
 Version: 2022.1 (Build 3526262 on Mon Apr 18 15:48:16 MDT)
 Project: Lab1
 Status: **Pass**

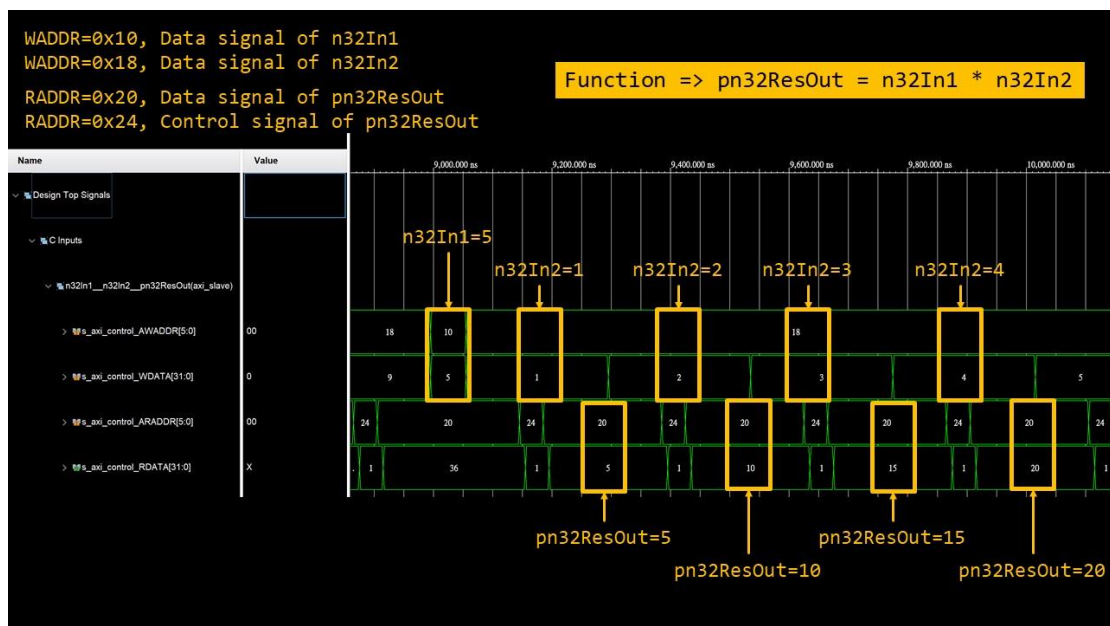
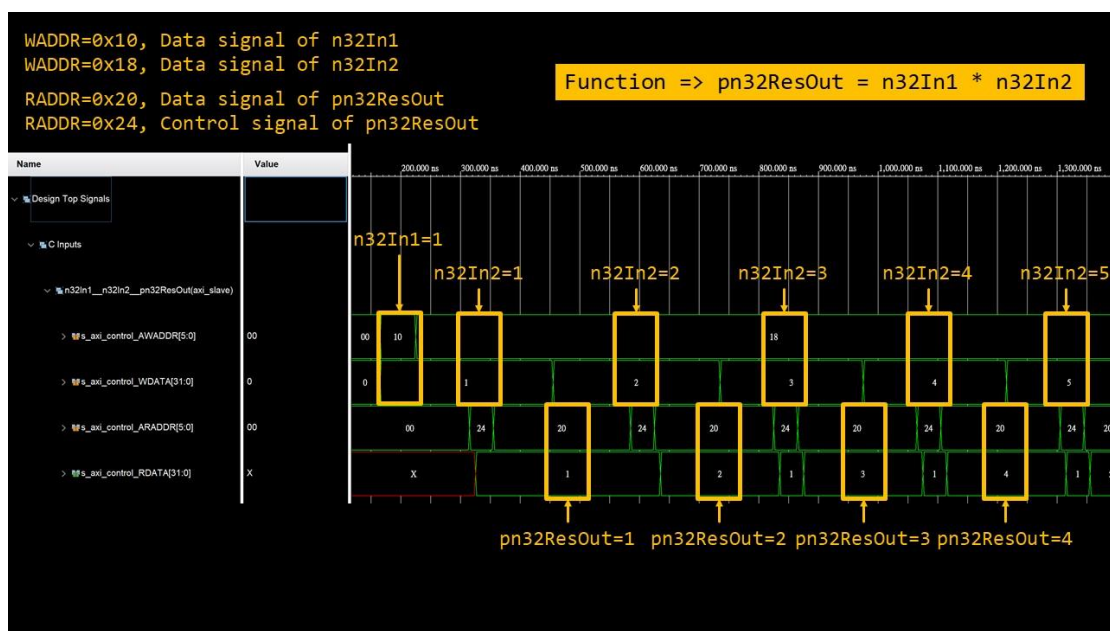
Solution: solution1 (Vivado IP Flow)
 Product: zynq
 Target device: xc7z020-clg400-1

Cosim Options

Tool: Vivado XSIM
 Dump Trace: all
 RTL: Verilog

Performance Estimates

Modules & Loops	Avg II	Max II	Min II	Avg Latency	Max Latency	Min Latency
● multip_2num 24	28	24	3	3	3	3



- Jupyter Notebook execution results

```
1 from __future__ import print_function
2
3 import sys, os
4
5 sys.path.append('/home/xilinx')
6 os.environ['XILINX_XRT'] = '/usr'
7 from pynq import Overlay
8
9 if __name__ == "__main__":
10     print("Entry:", sys.argv[0])
11     print("System argument(s):", len(sys.argv))
12
13     print("Start of \"" + sys.argv[0] + "\"")
14
15     ol = Overlay("/home/xilinx/jupyter_notebooks/Multip2Num.bit")
16     regIP = ol.multip_2num_0
17
18     for i in range(9):
19         print("=====")
20         for j in range(9):
21             regIP.write(0x10, i + 1)
22             regIP.write(0x18, j + 1)
23             Res = regIP.read(0x20)
24             print(str(i + 1) + " * " + str(j + 1) + " = " + str(Res))
25     print("=====")
26     print("Exit process")
```




```
1 Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
2 System argument(s): 3
3 Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
4 =====
5 1 * 1 = 1
6 1 * 2 = 2
7 1 * 3 = 3
8 1 * 4 = 4
9 1 * 5 = 5
10 1 * 6 = 6
11 1 * 7 = 7
12 1 * 8 = 8
13 1 * 9 = 9
14 =====
15 2 * 1 = 2
16 2 * 2 = 4
17 2 * 3 = 6
18 2 * 4 = 8
19 2 * 5 = 10
20 2 * 6 = 12
21 2 * 7 = 14
22 2 * 8 = 16
23 2 * 9 = 18
24 =====
25 3 * 1 = 3
26 3 * 2 = 6
27 3 * 3 = 9
28 3 * 4 = 12
29 3 * 5 = 15
30 3 * 6 = 18
31 3 * 7 = 21
32 3 * 8 = 24
33 3 * 9 = 27
34 =====
35 4 * 1 = 4
36 4 * 2 = 8
37 4 * 3 = 12
38 4 * 4 = 16
39 4 * 5 = 20
40 4 * 6 = 24
41 4 * 7 = 28
42 4 * 8 = 32
43 4 * 9 = 36
44 =====
45 5 * 1 = 5
46 5 * 2 = 10
47 5 * 3 = 15
48 5 * 4 = 20
49 5 * 5 = 25
50 5 * 6 = 30
51 5 * 7 = 35
52 5 * 8 = 40
53 5 * 9 = 45
```



```
1 =====
2 6 * 1 = 6
3 6 * 2 = 12
4 6 * 3 = 18
5 6 * 4 = 24
6 6 * 5 = 30
7 6 * 6 = 36
8 6 * 7 = 42
9 6 * 8 = 48
10 6 * 9 = 54
11 =====
12 7 * 1 = 7
13 7 * 2 = 14
14 7 * 3 = 21
15 7 * 4 = 28
16 7 * 5 = 35
17 7 * 6 = 42
18 7 * 7 = 49
19 7 * 8 = 56
20 7 * 9 = 63
21 =====
22 8 * 1 = 8
23 8 * 2 = 16
24 8 * 3 = 24
25 8 * 4 = 32
26 8 * 5 = 40
27 8 * 6 = 48
28 8 * 7 = 56
29 8 * 8 = 64
30 8 * 9 = 72
31 =====
32 9 * 1 = 9
33 9 * 2 = 18
34 9 * 3 = 27
35 9 * 4 = 36
36 9 * 5 = 45
37 9 * 6 = 54
38 9 * 7 = 63
39 9 * 8 = 72
40 9 * 9 = 81
41 =====
42 Exit process
```